Exercises

Exercise 5

Given the 3 key-value pairs $\{(\boldsymbol{k}_1, \boldsymbol{v}_1), (\boldsymbol{k}_2, \boldsymbol{v}_2), (\boldsymbol{k}_3, \boldsymbol{v}_3)\} = \{([2,0,1]^\top, [-1,1]^\top), ([0,1,0]^\top, [-1,0]^\top), ([0,2,0]^\top, [0,1]^\top)\}$, and the query $\boldsymbol{q} = [1,2,3]^\top$, compute the output of attention pooling for additive attention, and scale dot-product attention, respectively. The necessary weights are generated randomly having values from the set $\{-1,0,1\}$ (you can generate them yourself, not all zeros).

Solution:

- we have the following formulas:

$$f(\boldsymbol{q}, (\boldsymbol{k}_1, \boldsymbol{v}_1), \ldots, (\boldsymbol{k}_m, \boldsymbol{v}_m)) = \sum_{i=1}^{m} \alpha(\boldsymbol{q}, \boldsymbol{k}_i)\boldsymbol{v}_i \in \mathbb{R}^v,$$

$$\alpha(\boldsymbol{q}, \boldsymbol{k}_i) = \text{softmax}(a(\boldsymbol{q}, \boldsymbol{k}_i)) = \frac{\exp(a(\boldsymbol{q}, \boldsymbol{k}_i))}{\sum_{j=1}^{m} \exp(a(\boldsymbol{q}, \boldsymbol{k}_j))} \in \mathbb{R},$$

where the query $\boldsymbol{q} \in \mathbb{R}^q$, the m key-value pairs $(\boldsymbol{k}_1, \boldsymbol{v}_1), \ldots, (\boldsymbol{k}_m, \boldsymbol{v}_m)$, where any $\boldsymbol{k}_i \in \mathbb{R}^k$ and any $\boldsymbol{v}_i \in \mathbb{R}^v$

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- *additive attention*:

$$a(\boldsymbol{q}, \boldsymbol{k}) = \boldsymbol{w}_v^\top \tanh(\boldsymbol{W}_q \boldsymbol{q} + \boldsymbol{W}_k \boldsymbol{k}) \in \mathbb{R},$$

  where the parameters are $\boldsymbol{W}_q \in \mathbb{R}^{h \times q}$, $\boldsymbol{W}_k \in \mathbb{R}^{h \times k}$, and $\boldsymbol{w}_v \in \mathbb{R}^h$
- *scaled dot-product attention*:

$$a(\boldsymbol{q}, \boldsymbol{k}) = \boldsymbol{q}^\top \boldsymbol{k} / \sqrt{d} \in \mathbb{R},$$

  where $q = k = d$
- in our case, $m = 3$, $k = 3$, $v = 2$, $q = 3$
- we take $h = 2$ and

$$\boldsymbol{w}_v = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \in \mathbb{R}^2,$$

$$\boldsymbol{W}_q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \in \mathbb{R}^{2 \times 3},$$

$$\boldsymbol{W}_k = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \in \mathbb{R}^{2 \times 3}.$$

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- we apply the formula for additive attention:

$$
\begin{aligned}
a(\boldsymbol{q}, \boldsymbol{k}_1) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh\left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tanh 4 \\ \tanh 2 \end{bmatrix} \\
&= \tanh 4 \in \mathbb{R},
\end{aligned}
$$

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$$
\begin{aligned}
a(\boldsymbol{q}, \boldsymbol{k}_2) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh \left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh \left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tanh 5 \\ \tanh 2 \end{bmatrix} \\
&= \tanh 5 \in \mathbb{R},
\end{aligned}
$$

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$$
\begin{aligned}
a(\boldsymbol{q}, \boldsymbol{k}_3) &= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh \left(\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 0 \end{bmatrix} \tanh \left(\begin{bmatrix} 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} \right) \\
&= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \tanh 6 \\ \tanh 2 \end{bmatrix} \\
&= \tanh 6 \in \mathbb{R},
\end{aligned}
$$

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$$
\begin{aligned}
f(\boldsymbol{q}, (\boldsymbol{k}_1, \boldsymbol{v}_1), (\boldsymbol{k}_2, \boldsymbol{v}_2), (\boldsymbol{k}_3, \boldsymbol{v}_3)) &= \frac{\exp(\tanh 4)}{\exp(\tanh 4) + \exp(\tanh 5) + \exp(\tanh 6)} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
&\quad + \frac{\exp(\tanh 5)}{\exp(\tanh 4) + \exp(\tanh 5) + \exp(\tanh 6)} \begin{bmatrix} -1 \\ 0 \end{bmatrix} \\
&\quad + \frac{\exp(\tanh 6)}{\exp(\tanh 4) + \exp(\tanh 5) + \exp(\tanh 6)} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= \frac{1}{\exp(\tanh 4) + \exp(\tanh 5) + \exp(\tanh 6)} \\
&\quad \times \begin{bmatrix} -\exp(\tanh 4) - \exp(\tanh 5) \\ \exp(\tanh 4) + \exp(\tanh 6) \end{bmatrix} \in \mathbb{R}^2.
\end{aligned}
$$

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- now, we apply the formula for scaled dot-product attention:

$$a(\boldsymbol{q}, \boldsymbol{k}_1) = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \frac{5}{\sqrt{3}} \in \mathbb{R},$$

$$a(\boldsymbol{q}, \boldsymbol{k}_2) = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \frac{2}{\sqrt{3}} \in \mathbb{R},$$

$$a(\boldsymbol{q}, \boldsymbol{k}_3) = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} = \frac{4}{\sqrt{3}} \in \mathbb{R},$$

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$$
\begin{aligned}
f(\boldsymbol{q}, (\boldsymbol{k}_1, \boldsymbol{v}_1), (\boldsymbol{k}_2, \boldsymbol{v}_2), (\boldsymbol{k}_3, \boldsymbol{v}_3)) &= \frac{\exp\left(\frac{5}{\sqrt{3}}\right)}{\exp\left(\frac{5}{\sqrt{3}}\right) + \exp\left(\frac{2}{\sqrt{3}}\right) + \exp\left(\frac{4}{\sqrt{3}}\right)} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
&\quad + \frac{\exp\left(\frac{2}{\sqrt{3}}\right)}{\exp\left(\frac{5}{\sqrt{3}}\right) + \exp\left(\frac{2}{\sqrt{3}}\right) + \exp\left(\frac{4}{\sqrt{3}}\right)} \begin{bmatrix} -1 \\ 0 \end{bmatrix} \\
&\quad + \frac{\exp\left(\frac{4}{\sqrt{3}}\right)}{\exp\left(\frac{5}{\sqrt{3}}\right) + \exp\left(\frac{2}{\sqrt{3}}\right) + \exp\left(\frac{4}{\sqrt{3}}\right)} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
&= \frac{1}{\exp\left(\frac{5}{\sqrt{3}}\right) + \exp\left(\frac{2}{\sqrt{3}}\right) + \exp\left(\frac{4}{\sqrt{3}}\right)} \\
&\quad \times \begin{bmatrix} -\exp\left(\frac{5}{\sqrt{3}}\right) - \exp\left(\frac{2}{\sqrt{3}}\right) \\ \exp\left(\frac{5}{\sqrt{3}}\right) + \exp\left(\frac{4}{\sqrt{3}}\right) \end{bmatrix} \in \mathbb{R}^2.
\end{aligned}
$$